The database covers landslide-vulnerable regions of India in the Himalayas and Western Ghats. Satellite data of high to very high resolution nature such as IRS-1D PAN+LISS-III, Resourcesat-1, 2 and 2A LISS-IV Mx, Cartosat-1 and 2S, Aerial images were used in the mapping of landslides. Change detection using visual (manual) and digital (automatic) techniques were used to prepare the landslide inventory database. Some mapped landslides were validated in the field using mobile App and from the news report. The database mainly contains three types of inventory – seasonal, event-based and route-wise for the 2000-2017 period. Seasonal inventory contains pan-India landslide database corresponding to the 2014 and 2017 rainy season in India. The inventory database is available in a web GIS platform in the Bhuvan portal. The landslide inventory database shows the hotspot areas.

Table 2. Landslides inventory database of India.

Sl. No.	State/UTs	Monsoon season 2014	Monsoon season 2017	Field-based/year	Event-based / year	Total
1	**Jammu and Kashmir**	6826	19	434 / 2011	1 / 2015	**7280**
2	**Ladakh**	23		-	-	**23**
3	**Himachal Pradesh**	922	172	413 / 1998	1/ 2017 51/2013 2/2021	**1561**
4	**Uttarakhand**	1593	455	1419 / 1998	32/2003 307/2010 473 / 2012 6610 / 2013 1 / 2017 329/2021 1/2022	**11219**
5	**Sikkim**	73	79	-	1408 / 2011 8 / 2012 1/ 2016	**1569**
6	**West Bengal**	24	82	-	66 / 2011	**172**
7	**Arunachal Pradesh**	2904	4709	-	75 / 2016 1/2021	**7689**
8	**Nagaland**	54	2071	-	7/2017	**2132**
9	**Manipur**	379	4559	-	556/2017 1/2022	**5494**
10	**Mizoram**	1205	2254	-	8926/2017	**12385**
11	**Tripura**	56	8014	-	-	**8070**
12	**Assam**	1243	793	-	533/2017 5091/2022	**2569**
13	**Meghalaya**	2127	512	-	-	**2639**
14	**Maharashtra**	97	3	-	5012/2021	**5112**
15	**Goa**	2	1	-	-	**3**
16	**Karnataka**	82	19	-	993/2018	**1094**
17	**Kerala**	9	45	-	5191/2018 756/2019 09/2020 29/2021	**6039**
18	**Tamil Nadu**	79	8	-	603/2018	**690**
19	**Haryana**	-	100	-	-	**100**
Total		**17,698**	**23,895**	**2,266**	**37,074**	**80933**

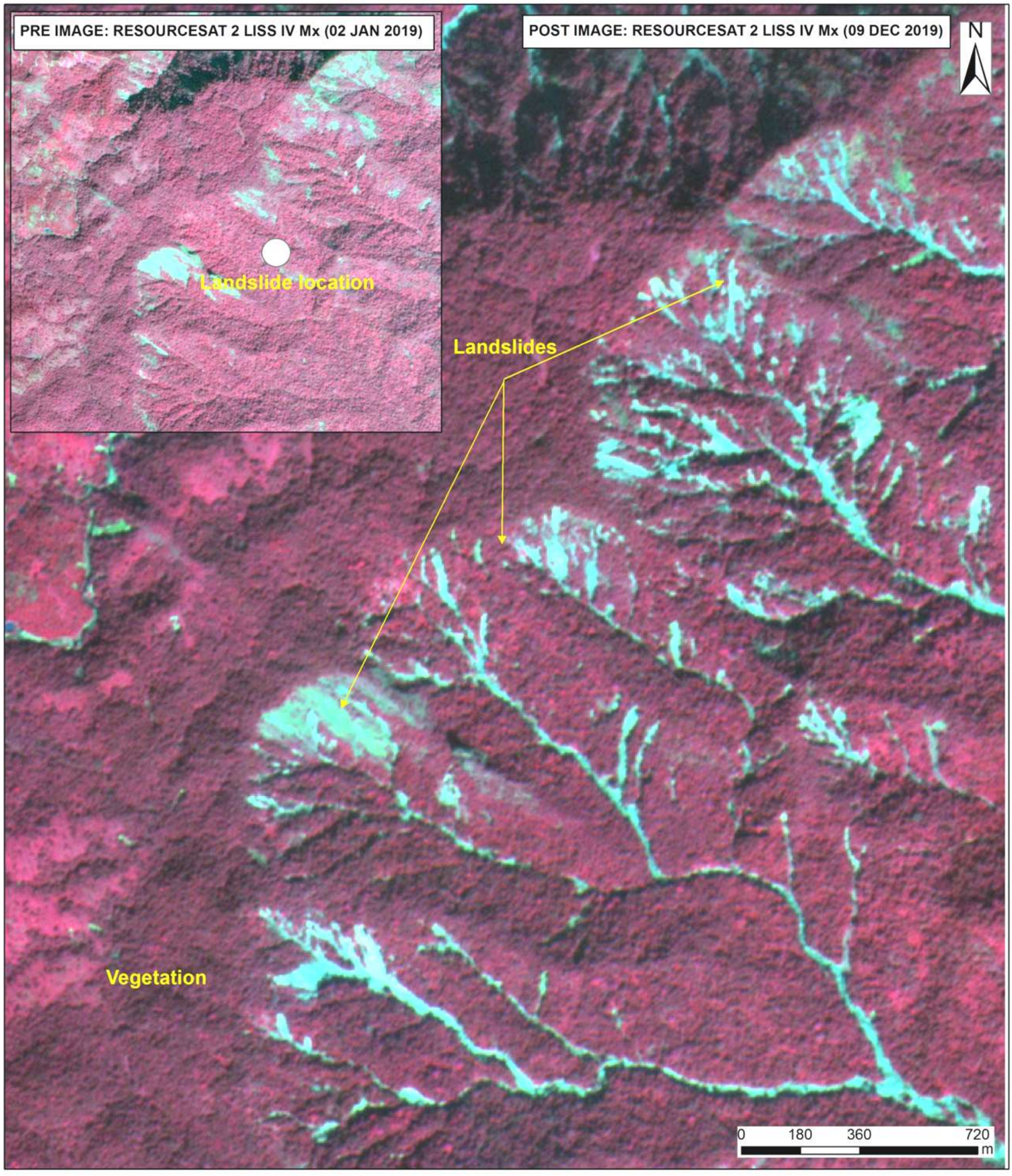


Figure 42. Landslide from Puthumala region, Wayanad district, Kerala. Most of these landslides are channelised debris flows

Debris Flow Modelling

Dynamic modelling of debris flow landslides has become an increasingly important practice for simulating the characteristics and behaviour of debris flow relying on the physical laws of conservation of mass, momentum and energy (Melo et al., 2018). This model approach can be broadly classified as the lumped mass moment where motion of slide is considered as a single point which is not always the case or based on continuum mechanics associated with a distinct rheology (Hungr, 1995). To simulate the deformation of the moving mass along the flow path, choosing appropriate friction parameters and material rheologies is taxing (Rickenmann, 2005).

Numerical debris flow modelling was carried out for several major landslides in Kerala to characterize the landslide geometry and assess their impact.

For modeling debris flow paths, Rapid Mass Movement Simulation (RAMMS) software developed by WSL Institute for Snow and Avalanche Research SLF and the Swiss Federal Institute for Forest, Snow and Landscape Research WSL, was used. The model gives flow height, velocity and pressure of debris flow. The main input parameters to be ingested in the model are high resolution Digital Elevation Model (DEM) for deriving topographic parameters, release area and friction information. CartoDEM (2.5 m) was used for flow modeling (Jain et al. 2021).

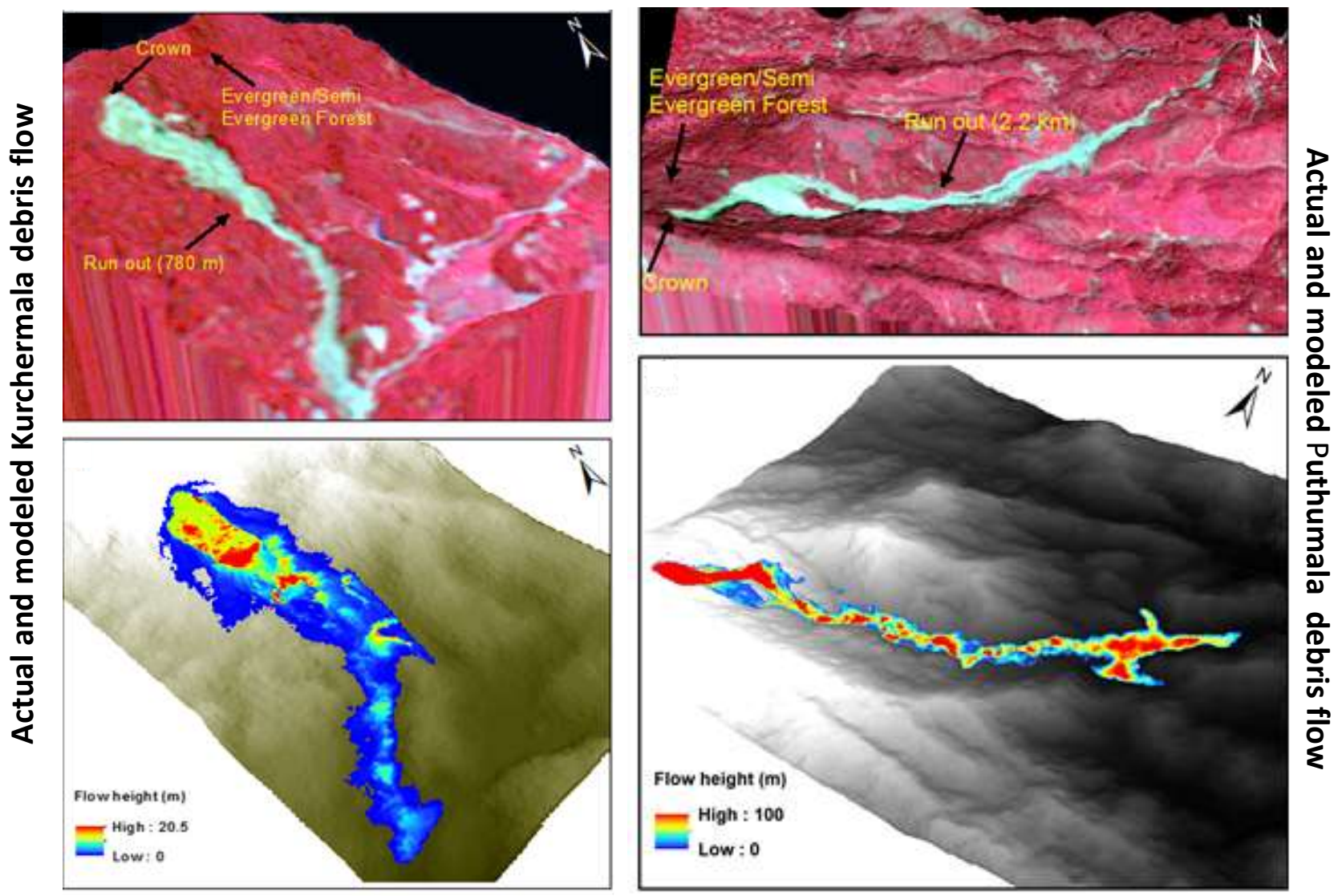


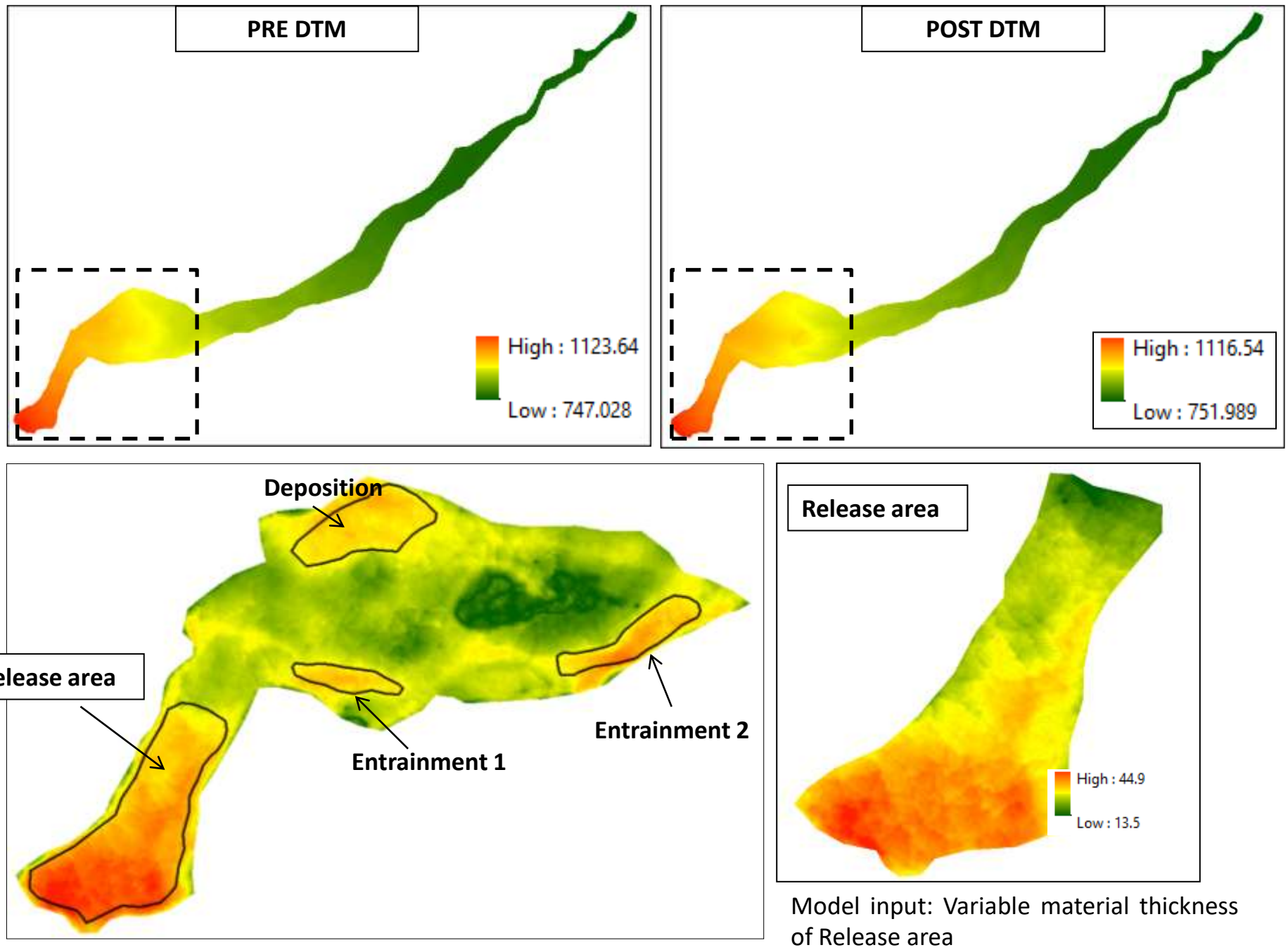
Figure 49. Flow modeling for Puthumala Landslide, Wayanad District, Kerala.

UAV data in landslide modeling

Photogrammetric techniques have been increasingly used because of their capability to rapidly reconstruct the 3-D topography from aerial photographs and provided such data exist for different time periods, allow objective change detection (Martha et al, 2010). Knowledge of failure volumes is also critical for a more accurate understanding of the landslide process and its flow, which shows the affected areas.

The increasing availability of high-resolution UAV data, quantitative studies on landform changes using DEMs have become a viable option.

UAV data were acquired over the Puthumala landslide in Kerala area on 22 March 2021. The resolution of orthoimage of UAV is 2 cm and the grid size of DTM is 8 cm. The positional accuracy of the image is 1-2 m. . Pre and post-DEM were utilized for volumetric estimation and release area mapping.



Model input: Variable material thickness of Release area

Figure 50. Landslide modeling using UAV data, Puthumala Landslide, Wayanad District, Kerala.